

Supplementary Digital Content Table 1. Donor, recipient, intraoperative, and perfusion characteristics of the HOPE-ECD-DBD trial*

Characteristics	All patients (n=46)	SCS (n=23)	HOPE (n=23)
Donor characteristics			
Age /median years [IQR]/	72 [59-78]	71 [59-78]	73 [60-78]
BMI /median kg/m ² [IQR]/	28 [25-31]	28 [24-31]	28 [26-31]
Sex Female : Male /n (%) /	21 (46%) : 25 (54%)	10 (44%) : 13 (56%)	11 (48%) : 12 (52%)
ET-DRI Score ^a	2.050 [1.878-2.218]	1.960 [1.830-2.240]	2.080 [1.910-2.210]
Cause of death /n (%) /			
Cerebrovascular accident	30 (65%)	15 (65%)	15 (65%)
Anoxia	10 (22%)	3 (13%)	7 (31%)
Trauma	6 (13%)	5 (22%)	1 (4%)
Donor Resuscitation/yes n (%) /	14 (31%)	6 (26%)	8 (35%)
Graft selection ECD criteria/n (%) ^b			
donor age≥65			
donor ICU≥7 days	31 (67%)	14 (61%)	17 (74%)
donor BMI>30 kg/m ²	13 (28%)	5 (22%)	8 (35%)
macro/mixed steatosis>40%	13 (28%)	7 (30%)	6 (26%)
serum sodium>165 mmol/L	10 (22%)	5 (22%)	5 (22%)
serum AST/ALT >3x of normal	3 (7%)	2 (9%)	1 (4%)
serum bilirubin>2 mg/dL	14 (30%)	8 (35%)	6 (26%)
	3 (7%)	1 (4%)	2 (9%)
Number of ECD criteria reached /n (%) /			
1 out of 7	13 (28%)	7 (30%)	6 (26%)
2-3 out of 7	26 (57%)	14 (61%)	12 (52%)
>3 out of 7	7 (15%)	2 (9%)	5 (22%)
Microsteatosis /n (%) ^c			
None to mild (0-29%)	32 (70%)	16 (70%)	16 (70%)
Moderate to severe (≥30%)	10 (22%)	6 (26%)	4 (17%)
Macrosteatosis /n (%) ^c			
None to mild (0-29%)	39 (85%)	21 (91%)	18 (78%)
Moderate to severe (≥30%)	3 (7%)	1 (4%)	2 (9%)
Graft weight /median g [IQR]/	1643 [1379-1980]	1619 [1346-1984]	1643 [1342-1878]
Recipient characteristics			
Age /median years [IQR]/	62 [55-65]	63 [56-67]	60 [52-64]
BMI /median kg/m ² [IQR]/	28 [25-31]	28 [26-34]	28 [23-31]
Sex Female : Male /n/	8 (17%) : 38 (83%)	3 (13%) : 20 (87%)	5 (22%) : 18 (78%)
labMELD /median [IQR]/	15 [9-25]	17 [8-25]	13 [9-18]
BAR Score /median [IQR]/ ^d	6 [4-10]	7 [4-11]	5 [3-8]
Listing indication /n (%) /			
Alcoholic cirrhosis	11 (24%)	7 (31%)	4 (17%)
Viral hepatitis	3 (7%)	0 (0%)	3 (13%)
HCC in cirrhosis	20 (43%)	9 (39%)	11 (48%)
PSC/PBC	2 (4%)	1 (4%)	1 (4%)
Other	10 (22%)	6 (26%)	4 (17%)
Pre-transplant abdominal surgery /n (%) /	9 (20%)	2 (9%)	7 (30%)
eGFR at admission /median mL/min/1.73m ² [IQR]/ ^e	89 [56-98]	79 [54-97]	91 [60-100]
Platelet counts at admission /median nL [IQR]/	98 [61-138]	98 [58-113]	97 [66-139]
Intraoperative and perfusion characteristics			
Total cold preservation time /median min [IQR]/	495 [465-593]	502 [467-584]	495 [447-612]
Cold preservation time before HOPE/median min [IQR]/	375 [311-465]	NA	375 [311-465]

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Machine perfusion time /median min [IQR]/	145 [101-203]	NA	145 [101-203]
Warm ischemic time /median min [IQR]/	43 [37-52]	45 [39-52]	39 [35-54]
Duration of surgery /median min [IQR]/	403 [324-474]	420 [308-495]	395 [329-452]
Intraoperative RBC /median units [IQR]/	4 [2-7]	4 [2-7]	3 [2-6]
Intraoperative FFP /median units [IQR]/	16 [10-24]	16 [10-18]	18 [9-26]

^aAdopted and modified with permission from Czigany et al. Annals of Surgery 2021; ^bBased on Braat et al. American Journal of Transplantation 2021; ^cBased on Czigany et al. BMJ Open 2017 and the definitions of ECD criteria liver allografts in DBD donors as recommended by the Germany Medical Chamber (Bundesärztekammer) ^dBased on McCormack et al. Annals of Surgery 2007; ^eBased on Dutkowski et al. Annals of Surgery 2011; ^fBased on the CKD-EPI (Chronic Kidney Disease Epidemiology Collaboration) equation, vdBrand et al. Nephrology Dialysis Transplantation 2011.

Abbreviations used: SCS, static cold storage; HOPE, hypothermic oxygenated machine perfusion; IQR, interquartile range; BMI, body mass index; ET-DRI, Eurotransplant donor risk index; ECD, extended criteria donor; ICU, intensive care unit; AST, aspartate aminotransferase; ALT, alanine aminotransferase; MELD, model for end-stage liver disease; BAR, balance of risk; HCC, hepatocellular carcinoma; PSC, primary sclerosing cholangitis; PBC, primary biliary cholangitis; eGFR, estimated glomerular filtration rate; CKD-EPI, Chronic Kidney Disease Epidemiology Collaboration; RBC, packed red blood cells; FFP, fresh frozen plasma; NA, not applicable